

Racing Against Tomorrow's Safeguards

Is the AI race really an arms race?

Pavel Kocourek | pkocourek.com | mentored by Katja Grace (AI Impacts)

Abstract

The AI race is widely called an **arms race** — a prisoner's dilemma where every lab wants to race.

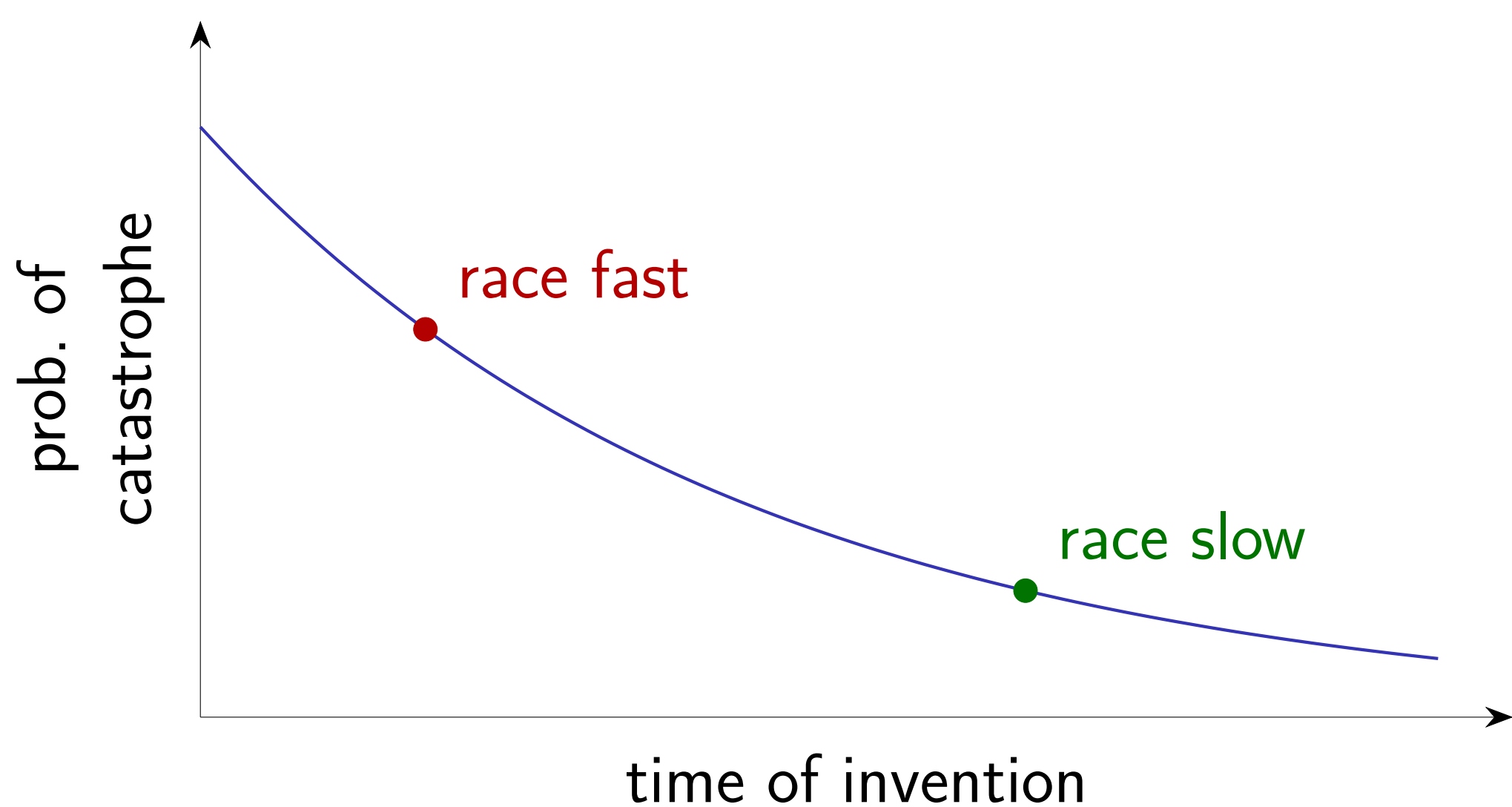
But the world keeps getting more prepared, so the risk of catastrophe *falls over time*.

In a Loury (1979) race with that falling risk, a self-interested lab **holds back on its own** — and a calm slowdown is an equilibrium

Introduction

Katja Grace argues, informally, that “*AI Is Not an Arms Race*” (TIME, 2023).

Does game theory support that?



Society keeps getting readier for a *fixed* AI capability — so catastrophe risk **falls over time**.

Racing faster pulls invention into the **high-risk window**; waiting lets it arrive safer

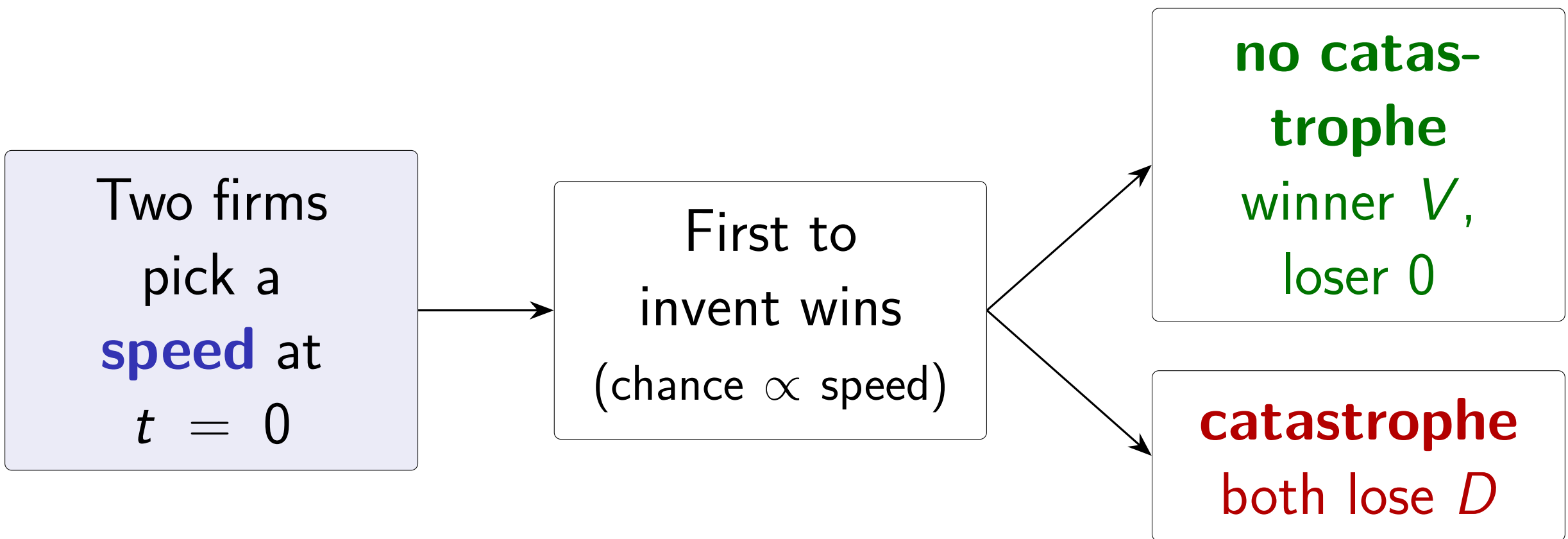
Prior work holds risk *fixed*; this paper lets it fall over time.

Grace 2022 · Hendrycks et al. 2023 · Armstrong–Bostrom–Shulman 2016 · Jones 2024 ·

Tan 2025

The model

Two firms each pick a development **speed** at $t = 0$; the first to invent wins.



A Loury (1979) patent race — with a chance of catastrophe added; effort is free.

As the world prepares, the risk decays:

$$p(t) = p_0 e^{-\delta t}$$

Nash equilibrium: neither firm can do better by changing only its own speed

Result 1 — risk starts certain ($p_0 = 1$)

Even with *free* effort, firms hold back **voluntarily**:

$$\lambda^* = \frac{1}{2} \frac{V\delta}{D}$$

The whole prize is competed away — not by effort, but by the catastrophe risk hurrying creates:

$$U^* = 0$$

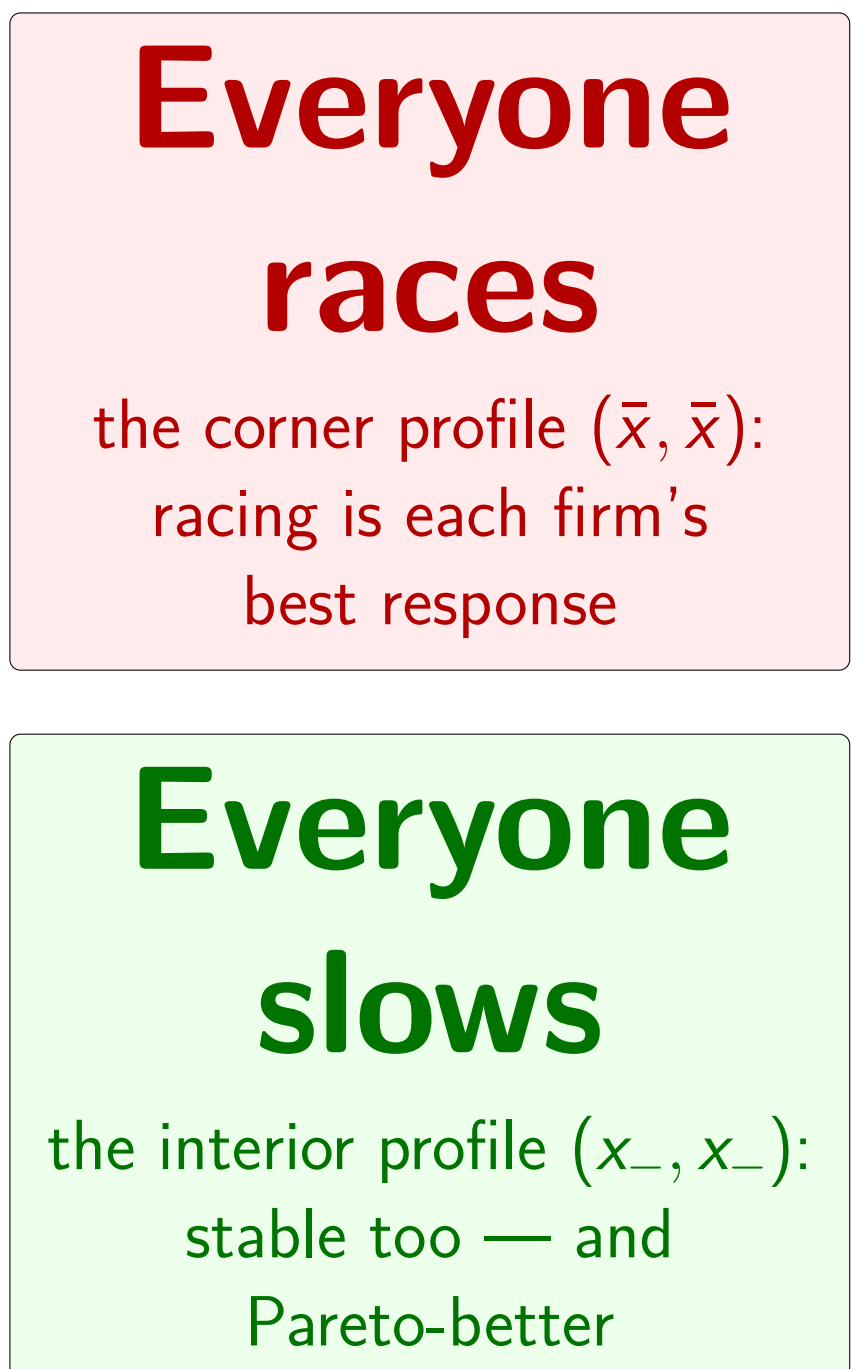
Preparedness treadmill — equilibrium catastrophe probability

$$\frac{V}{V + 2D}, \quad \text{independent of } \delta$$

faster safety progress is absorbed by faster racing

Result 2 — the realistic case ($p_0 \in (0, 1)$)

Catastrophe is likely, but *not* certain, at the start — now the game has **two symmetric equilibria**:



A calmer slowdown equilibrium exists whenever $p_0 > p(r) = 1 - \frac{1}{(1+r)^2}$, $r = V/D$, and it **Pareto-dominates** racing

Discussion — policy

Calling the AI race an arms race is a **self-fulfilling prophecy**: it coordinates beliefs on the bad equilibrium both labs would gladly leave.

The binding problem is **coordination**, not unilateral restraint.

A modest cap on development speed can tip firms onto the good equilibrium

References

- Loury, G. (1979). Market structure & innovation. *QJE*
- Grace, K. (2022). *Let's think about slowing down AI*
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- Hendrycks et al. (2023)
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- Jones, C. (2024)
- Tan (2025). *The Suicide Region*

Working paper — full references in the report